Diagnostic Features

- Presence of multilayered epidermis with thick cuticle.
- Sunken stomata are present only in the lower epidermis.
- Mesophyll is well differentiated into palisade and spongy parenchyma.
- Mechanical tissues are well developed.

II - Fresh or Preserved Specimens

Exercise 4: Natural methods of Vegetative Propagation in Plants

Aim: To study and identify the types of natural methods of vegetative propagation in plants.

Principle: Natural vegetative reproduction is a form of asexual reproduction in which vegetative bud grows and develops into a new plant.

Requirements: Fresh / preserved specimens of *Zingiber*, *Chrysanthemum*, *Bryophullum*.

Ask the students to visit the nearest vegetable market and classify the vegetable into root, stem or leaf based on their utility and identify how many of them can be propagated through vegetative methods.

4 A. Vegetative Propagation by underground stem – Rhizome

Diagnostic Features

- Ginger is a underground stem which is called as Rhizome.
- Rhizomes are horizontal and swollen due to the storage of food materials.
- The terminal buds turn upwards to produce the aerial flowering shoot and the lateral buds grow out to form new rhizomes.

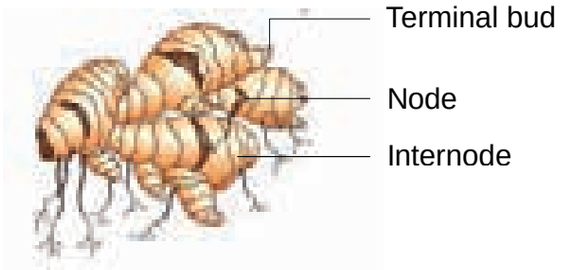


Figure 4a: Rhizome – Ginger

4 B. Vegetative Propagation by sub-aerial stem – Sucker

Diagnostic Features

- The suckers of *Chrysanthemum* are used for propagating plants.
- Suckers grows horizontally under the soil and then emerge out obliquely from the soil and give rise to a new plant or leafy shoot.
- The sucker has nodes and internodes. In the nodal region, it bears axillary buds above and adventitious roots below.

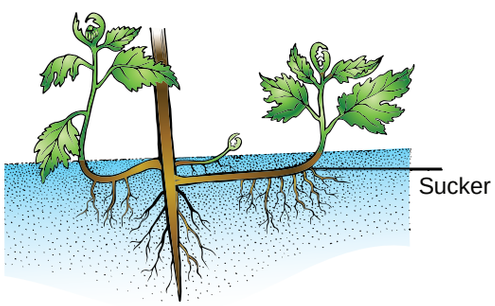


Figure 4b: Sucker - *Chrysanthemum*

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